Fail-closed class dispatch

Class-admitted work, data-plane bind. This machine has no leftover-hop edge.

Roque Briceño · Version 0.8.0 · 30 August 2026 · CC BY 4.0 · Reference implementation 0.5.0.

Abstract

A specialized model fleet is a checkable claim only if a work item of class c cannot complete a stage on a specialist or model outside the policy for c. Fail-closed class dispatch is that check: a work item carries a class and a frozen body; a policy gives each class an allow set $\pi(c)$ and a deny set $\delta(c)$; and each specialist a is bound to one API model identity $\phi(a)$. A stage may run only after a well-formed admit of some $a \in \pi(c) \setminus \delta(c)$, and only with the executed model recorded by Observe; if that bind cannot be served, the stage fails closed and the failure is published. Retry, where there is one, is the same item, the same class, a specialist not yet tried.

This is Clark–Wilson integrity applied to LLM binds, not a new access-control algebra: allow/deny, fail-safe defaults and dual control are old. The added obligations are three — ϕ is the identity Observe recorded from the inference client, leftover fallback is not an edge in this table, and the event contract is the witness set. An accidental hop is in scope; an adversarial worker that forges Observe is not.

Safety invariants I1–I6, I8 and I9 hold on the abstract stage machine under A0–A9; I7 bounds the Admit count; and under A10–A13, I10–I17 prove FCD-owned work/envelope pinning, package and receipt binding, steering scope, attempt-local cache identity and accepted-only memory promotion. Item liveness and model quality are not proved.

1. Problem

Routing, cascades and mixture-of-agents optimize score or spend (Chen et al., 2023; Ong et al., 2024; Wang et al., 2024); mixture-of-experts is a learned gate inside one network (Jacobs et al., 1991; Shazeer et al., 2017); orchestration says who calls whom. None of them forbids a hop to an unbound model when the bound provider is down, and none of them treats a model that refuses a class as illegal rather than low-scoring.

The sentence “we run specialists” is then unfalsifiable: the control plane can name one model while the data plane calls another; a 403 can look like work still in progress; a reviewer can be the author. The required property is class integrity — a pass record lies in $\pi(c) \setminus \delta(c)$ and uses $\phi(a)$.

2. Objects

A **work item** has class c, charge, frozen body hash, author set, required-stage list Required(c), and status open | failed | accepted. $\pi(c)$ and $\delta(c)$ are disjoint; on a check stage the effective allow set is $\pi\_chk(c, authors) = \pi(c) \setminus authors$. ϕ: A $\to$ M maps a specialist to an API identity, and display strings are compared after norm. u(m)=0 iff bind time returns 401, 403, 404, 429, exhausted, or not_found, and tried is the set of specialists already admitted on this stage.

A stage is well-formed only if a control event names class, specialist, declared model and body hash; prose that mentions a name is not a stage. Fail closed means publishing a fail_closed decision, after which the next move is ask, retry in $\pi(c) \setminus \delta(c) \setminus tried$, or stop. An **accepted artifact** exists only when every stage in Required(c) has passed, and the store accepts only accepted artifacts.

3. Process

1. Open a well-formed work item.

2. Admit a $\in \pi^*(c)\backslash\delta(c)\backslash$tried. If none, fail closed.

3. Bind. Writes m_decl = $\phi(a)$. If $u(\phi(a))=0$, fail closed. Does not write m_exec.

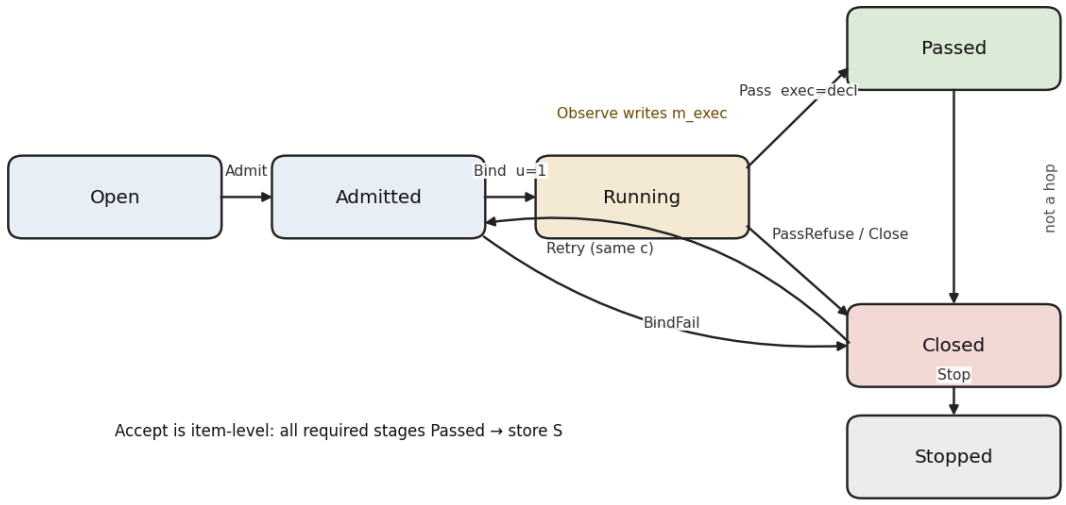
4. Observe. Copies m_exec from the provider call (A1).

5. Pass only if norm(m_exec)=norm(m_decl); else PassRefuse (F1).

6. When Required(c) is covered, Accept into the store.

A death watchdog outside the worker records death while Running (A2, A9). Close publishes.

Figure 1. Stage automaton (Observe machine)



4. What holds

Bind writes the declared identity only and Observe writes the executed one, so I1 is non-vacuous; under A0–A9, I1–I6, I8 and I9 are inductive, and I7 bounds the Admit count. Ask may idle. The leftover-hop picture is an illustration, not a corollary.

Figure 2. Why I1 is non-vacuous: two writers

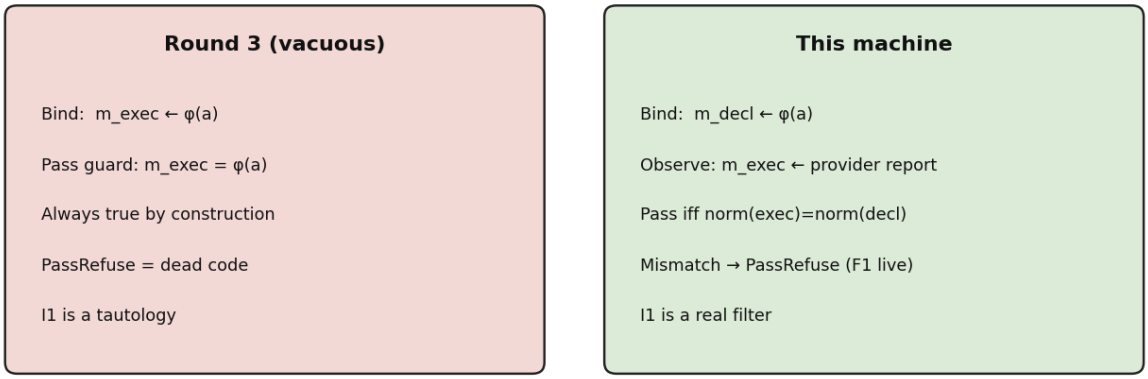
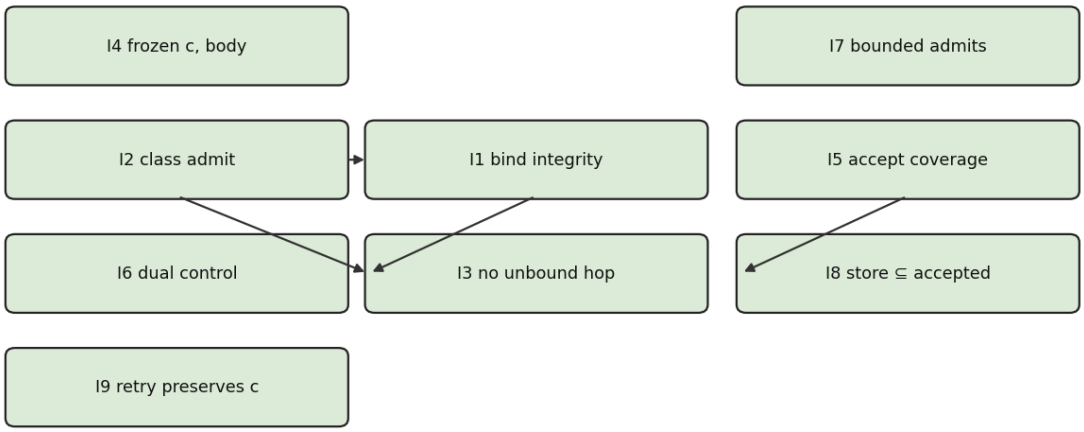


Figure 3. Stage theorem map (I1–I9)



Not stage theorems: quality, item liveness, provider physics, physical context isolation.

Id	Claim	Status
I1	Passed ⇒ norm(exec)=norm(decl)=norm(ϕ(a))	Theorem (Pass-time)
I2	Running/Passed ⇒ a ∈ π*\δ	Theorem
I3	No Pass with exec outside ϕ(π*\δ)	Theorem (via I1, I2)
I4	c and body frozen after Open	Theorem
I5	accepted ⇒ every required stage Passed	Theorem
I6	check Admit ⇒ a ∉ authors	Theorem
I7	≤ |π*\δ| Admits per stage	Bound, not termination

Id	Claim	Status
I8	id ∈ S ⇒ accepted	Theorem
I9	Retry does not write c	Theorem

4.1 Inductive proofs (condensed)

I1. Only Pass sets pc=Passed, and only when m_exec ≠ none and norm(m_exec)=norm(m_decl). Bind is the only writer of m_decl and writes ϕ(a). Observe may set a mismatch; then PassRefuse fires and Passed is unreachable. If Bind wrote m_exec, the guard would be tautological.

I2. a is written only by Admit, whose guard is a ∈ π*\δ\tried. Stages of one item are sequential (A4), so π* is the Admit-time snapshot.

I3. I1 gives exec = ϕ(a). I2 gives a ∈ π*\δ.

I4. No transition after Open writes c or body.

I5. Accept is the only writer of status=accepted and requires all required Passed.

I6. Check admit uses π_chk = π(c)\authors (A6) on the Admit-time snapshot.

I7. Each Admit adds a unused specialist to tried (A7). The allow set is finite (A0). Ask may idle; this is a bound on Admit count, not liveness.

I8. Accept is the only writer of store S and sets accepted in the same step (A8).

I9. Retry does not write c (I4).

5. Context, memory and cache envelope

A work item pins accepted project P and memory K at Open. Each gate Admit creates a monotonic attempt counter, unpredictable nonce and immutable base envelope containing agent/profile, exact provider/API model, instruction hash, context policy, tool authority, FCD cache identity and pre-Admit steering hash. Live steering advances a separate ordered continuation chain.

FCD builds canonical package bytes from include minus exclude, and the adapter independently hashes the bytes it submits; Pass requires current attempt/nonce, package, executor/run, latest steering and executed-model receipts. fresh_blind excludes author context and forbids executor continuity, and existing executor tools, sessions and provider caches remain external.

Id	Context-envelope claim
I10	Work P/K and base envelope are pinned
I11	Pass requires current package receipt
I12	fresh_blind manifest excludes author context
I13	Only serialized accepted CAS promotes memory
I14	No silent project/memory drift
I15	Steering stays in scope and latest ack is required
I16	FCD cache is full-identity attempt-local
I17	Prior-attempt receipts cannot authorize Pass

Theorems cover FCD manifests, receipts and state transitions. Adapter honesty, physical prompt isolation, hidden session residue, provider cache neutrality, model quality and impact-review correctness remain explicit assumptions or limits.

6. Faults

Each fault is a forbidden transition, not a metaphor.

Id	Forbidden step
F1	Pass with norm(m_exec) $\neq$ norm(ϕ(a))
F2	Two specialists, one runtime instance (needs instance field)
F3	After u=0, another call without fail-closed, or outside unused allow
F4	Running exit with no published close
F5	ϕ(a) not an API identity
F6	Pass with a $\in$ δ(c)
F7	Check admit with a $\in$ authors
F8	Run without a well-formed stage
F9	Stop in chat with status not accepted
F10	Same id, class changes; or retry of a $\in$ tried

Weight-sharing across specialists is extra config. It is not in I6.

7. Measurement

A named cut is [t0, t1], with W the class p95 of completed stage durations in the previous cut, or 12 minutes if n<30; exclude ts > t1–W, and evaluate π, δ, ϕ as-of event ts.

Figure 4. Measurement sample spaces (empty of numbers)

Rate	Sample space	Event	Pairing
Misbind	first Observe / stage	norm(exec)≠norm(decl)	with silent-fail
Silent fail	well-formed stages + orphan open	no decide/accept in W	fail-closed = published
Bleed	stages of class c	a ∉ π*(c) as-of ts	pair with silent-fail
Time-to-stage	well-formed stages	survival to decide/accept	not a mean

No empirical bars. Zeros are proofs only under write-ahead + total call/decide.

Zeros on misbind, bleed and silent-fail are a proof that those faults did not fire only if stage is write-ahead and call/decide are total; otherwise they are estimates biased clean. No numbers in this paper.

8. Reference implementation

The control plane is this machine, not a chat host or sessions sidebar. A verified project comes before intake. The operator may choose the gate agent, execution adapter, exact model and context policy before Admit. Agent instructions and exact-route readiness are visible; an unavailable route cannot start. Admit locks the envelope. Existing executors keep their mature tools and session stores; adapters return evidence and receipts but cannot Pass, Accept or promote memory.

The three-pane surface keeps project/capability atlas, selected work line plus bounded gate tray, and real artifact visible together. Questions stay anchored. Steering has explicit project, work, gate, stage, artifact and evidence scope. A pixel is trustworthy only when it projects an event or receipt.

9. Related work

Clark and Wilson (1987) already have constrained data items, transformation procedures, a certification relation, integrity verification and mandatory separation of duty: a work item is a CDI, a bound specialist is a TP, and Accept is a validated write.

Saltzer and Schroeder (1975) name fail-safe defaults; deny-override is standard MAC/RBAC; Thomas and Sandhu (1997) bind permission to a task instance. F1/F2/F5 are TOCTOU / control-plane versus data-plane divergence, and F7 is also LLM-as-judge self-preference.

MoE, MoA, FrugalGPT, RouteLLM and MoMA optimize a different objective, and orchestrator-worker (Anthropic, 2024) is a topology. Topology is not the process.

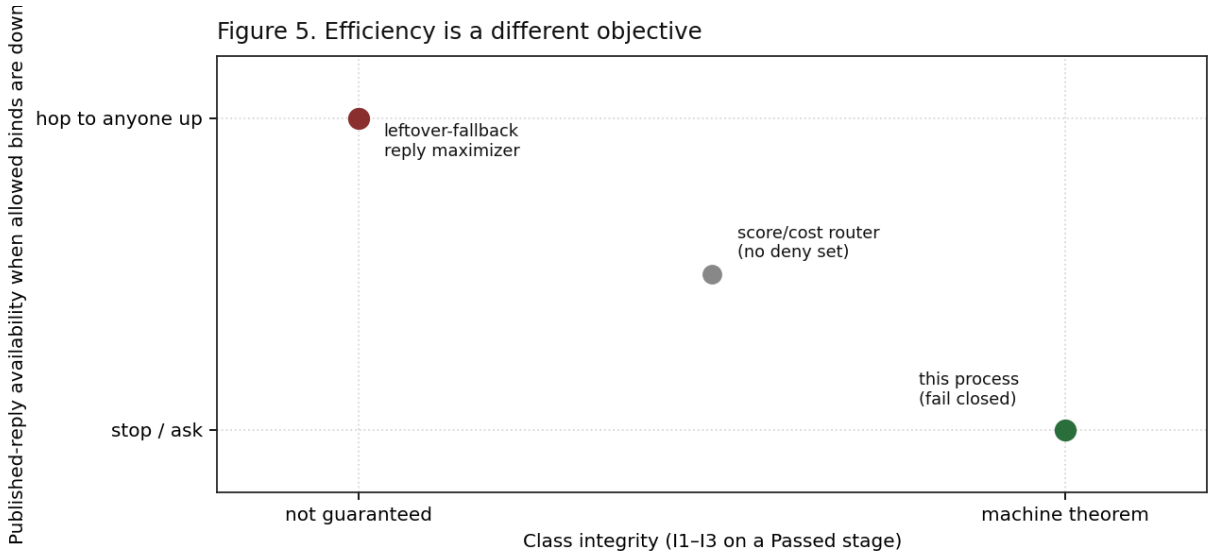
10. Limits

There is no dataset here, no quality theorem, and no item liveness. I1 is the Pass-time report, not execution history, and I1/I3 hold up to norm collision unless norm is injective; I10–I17 prove FCD-owned manifest, receipt, cache and promotion transitions. They do not prove adapter honesty, physical prompt isolation, hidden executor residue, provider cache neutrality or impact-review correctness.

10.1 Efficiency without sacrificing the theorems

Efficiency (tokens, latency, cache hits, published-reply rate) is a different objective from class integrity. A leftover-fallback reply-maximizer improves availability exactly when every allowed bind is down — the same state where this process is already not live. That is not a free lunch. It is a hop.

Figure 5. Efficiency is a different objective

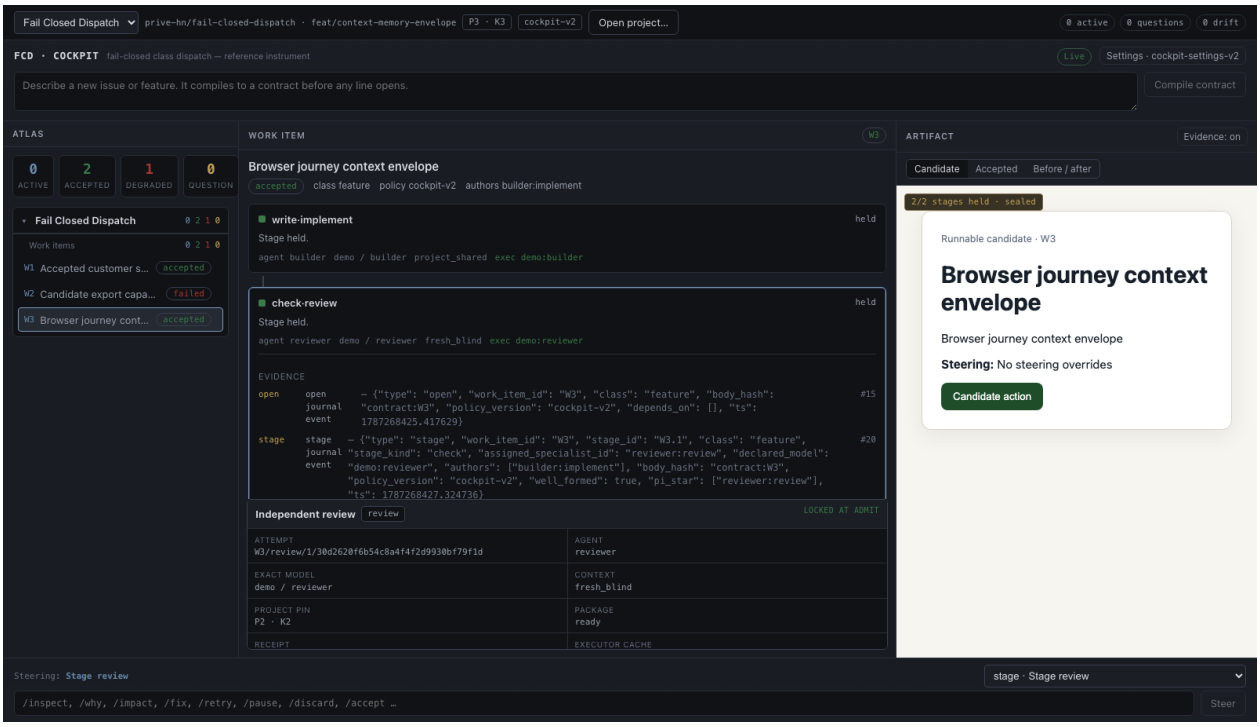


Safe efficiency: (1) FCD cache stays attempt-local and clears on Admit/Close; (2) mature executors may report provider prefix/session reuse, but it has no Pass authority; (3) parallelize only across distinct work items; (4) retry only in π*\δ\tried, bounded by I7; (5) use cheaper models only when they are explicitly selected in policy; (6) Ask instead of hopping when the remainder is empty. These choices preserve I1–I17 under the stated assumptions.

Unsafe efficiency: shared FCD runtime across specialists (F2), session restore that ignores the pinned envelope, cross-attempt FCD cache reuse, leftover fallback, or treating executor-reported cache telemetry as semantic context proof.

11. Interactive evidence layer

The reference implementation keeps verified project/capability atlas, every sibling work line, a bounded editable/locked gate tray and real artifact visible together. Questions stay anchored. Drift review and explicit multi-scope steering remain inside the same cockpit.



The cockpit does not widen the theorem: execution adapters preserve mature worker tools/sessions but cannot Pass, Accept or promote memory; skins are read-only projections; fcd remains the only writer of accepted state. The shown Chrome journey produced package/model receipts before acceptance.

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